A spectral Wigner–Eckart evaluation of the polyatomic Boltzmann collision operator with the DSMC frozen/non-frozen collision kernel and its internal-energy-modulated extension

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Abstract

We implement and validate the variable-hard-sphere (VHS) Borgnakke–Larsen collision kernel used by the direct simulation Monte Carlo (DSMC) community [Djordjić, Oblapenko, Pavić-Čolić & Torrilhon, *Continuum Mech. Thermodyn.* **35** (2023) 103–119] inside a spectral Wigner–Eckart factorization of the polyatomic Boltzmann collision operator. The kernel is a convex combination of a *non-frozen* channel (full Borgnakke–Larsen internal-energy redistribution) and a *frozen* channel (elastic, internal-energy preserving). We show that the frozen channel must be integrated in internal-energy space with no additional partition measure – a subtle point whose violation injects a spurious δ-dependence into the internal heat-flux relaxation rate. With the corrected channel the operator reproduces the Eucken Prandtl number at the analytic anchor to machine precision (2×10^{-14}) and the exact monatomic shear sector $P_\sigma^{(0)}/P_q^{(0)} = 3/2$ for all ζ. An independent Monte-Carlo evaluation of the elastic collision bracket confirms our spectral frozen Prandtl number to 3–4 digits and shows that it *differs* from the paper's analytic frozen formula at $\zeta \neq 0$, i.e. the spectral quadrature is the more accurate of the two. Finally we implement the internal-energy-modulated extended kernel and demonstrate that, after re-fitting the free parameters in our (more accurate) operator, it reaches the experimental Prandtl number and the bulk/shear viscosity ratio simultaneously – exactly and with admissible parameters for N_2 and CO, and exactly but essentially on the kernel-positivity boundary for the extreme case H_2.

Contents

1 Introduction

The Boltzmann collision operator for a polyatomic gas with a continuous internal-energy variable governs the relaxation of the distribution function $f(\mathbf{v}, I)$ towards equilibrium and fixes the gas transport coefficients (shear viscosity μ, bulk viscosity ν, thermal conductivity κ, and hence the Prandtl number Pr). Evaluating this operator on a spectral basis is expensive because the gain term is a high-dimensional integral over the collision geometry and the internal-energy redistribution.

The *Wigner–Eckart factorization* reduces this cost dramatically: rotation invariance of the collision kernel lets the operator be written as a contraction of a sparse, geometry-only *Gaunt* angular tensor with a dense, kernel-dependent *physical* tensor (here denoted the R-tensor). Only the R-tensor depends on the collision kernel, so a new kernel model requires changes only to its quadrature – the angular machinery is untouched.

This report integrates the DSMC collision-kernel model of Djordjić *et al.* [1] into such a spectral code. The model is the variable-hard-sphere Borgnakke–Larsen (VHS–BL) kernel, written as a convex combination of a *non-frozen* channel (internal energy redistributed) and a *frozen* channel (elastic, internal energy preserved). We (i) derive the correct spectral treatment of the frozen channel, (ii) validate the resulting transport coefficients against analytic anchors and against an independent Monte-Carlo computation, finding that the spectral operator is more accurate than the paper's analytic transport formulas at hard-potential exponents, and (iii) implement the internal-energy-modulated extended kernel and show it reaches experimental data.

2 The collision operator and the DSMC kernel

2.1 Operator and Wigner–Eckart factorization

The space-homogeneous collision operator is

$$Q(f,f)(\mathbf{v},I) = \int \left[f(\mathbf{v}',I')f(\mathbf{v}'_*,I'_*) - f(\mathbf{v},I)f(\mathbf{v}_*,I_*)\right] \mathcal{B}\, d\sigma\, dR\, dr\, d\mathbf{v}_*\, dI_*, \tag{1}$$

with the Borgnakke–Larsen parametrization of the post-collision state by the kinetic-energy partition $R \in [0,1]$ and the internal-energy partition $r \in [0,1]$:

$$e'_{\mathrm{tr}} = R, \qquad i' = r(1-R), \qquad i'_* = (1-r)(1-R), \tag{2}$$

where $i = I/E$, $i_* = I_*/E$ and $E = \frac{1}{2}|\mathbf{u}|^2 + I + I_*$ is the (conserved) total energy in code units. Expanding f on the spectral basis $\phi_{k\ell m,n}(\mathbf{v}, I) = R_{k\ell}(|\mathbf{v}|)Y_{\ell m}(\hat{\mathbf{v}})\, H_n(I)$ (associated-Laguerre $\times$ spherical-harmonic $\times$ internal-Laguerre) and using rotation invariance, Q factorizes into a sparse Gaunt tensor contracted with the dense R-tensor $R[k_1k_2k_3, n_1n_2n_3, \ell]$, evaluated by sum factorization (a Fubini cascade over the energy, kinetic-partition, internal-partition and scattering integrals).

2.2 The DSMC convex-split kernel

The DSMC kernel [1, eqs. (53)–(55)] is

$$\mathcal{B}_{\mathrm{DSMC}} = |\mathbf{u}|^\zeta \Big[\omega\, K_\delta\, R^{\zeta/2} \;+\; (1-\omega)\,\delta_{r-r'}\delta_{R-R'}\, e_{\mathrm{tr}}^{\zeta/2}\Big], \tag{3}$$

where $\zeta = 2(1-s_{\mathrm{visc}})$ is the relative-speed exponent (s_{visc} is the viscosity temperature exponent), δ is the number of internal degrees of freedom ($\nu = \delta/2 - 1$ is the internal-Laguerre parameter), $\omega = p_{\mathrm{int}}$ is the inelastic-collision probability, and $K_\delta = 2\Gamma(\delta + \frac{3}{2})/[\sqrt{\pi}\,\Gamma(\delta/2)^2]$ is the monatomic-consistency constant. The first term is the *non-frozen* (Borgnakke–Larsen) channel; the second is the *frozen* (elastic, $I' = I$, $I'_* = I_*$, $|\mathbf{u}'| = |\mathbf{u}|$) channel. Because the operator is linear in $\mathcal{B}$, the R-tensor is a convex sum, $R_{\mathrm{tot}} = \omega\, K_\delta R_{\mathrm{nf}} + (1-\omega)\, C^f_{\mathrm{VHS}} R_{\mathrm{fr}}$.

2.3 Spectral treatment of the two channels

Non-frozen channel. The single separable term $K_\delta |\mathbf{u}|^\zeta R^{\zeta/2}$ folds cleanly into the existing quadrature: $|\mathbf{u}|^\zeta \propto z^{\zeta/2}$ ($z = \frac{1}{2}|\mathbf{u}|^2$) is desingularized by the Gauss–Laguerre($\zeta/2$) energy rule, while $R^{\zeta/2}$ shifts the kinetic-partition Gauss–Jacobi weight to Jacobi$(2\nu+1, \frac{1}{2} + \frac{\zeta}{2})$. The integrand is polynomial after folding, so convergence is *spectral*.

Frozen channel (a subtle correction). The frozen collision is elastic with the internal energies as spectators. The natural and correct spectral treatment integrates the frozen channel *directly in internal-energy space* over the Gauss–Laguerre nodes (I, I_*) with weights $W_I W_{I_*}$; basis orthonormality $\int H_a H_b\, I^\nu e^{-I}\, dI = \delta_{ab}$ then carries the frozen Kronecker structure $I' = I$. There is *no* (R, r)-partition phase-space measure in this channel.

A tempting but incorrect alternative collapses the Dirac deltas $\delta_{r-r'}\delta_{R-R'}$ against the Borgnakke–Larsen measure $H_\delta(r, R) = (r(1-r))^\nu (1-R)^{2\nu+1}\sqrt{R}$ and leaves H_δ as an explicit weight. Because the frozen routine already integrates over the internal energies, this double-counts a non-polynomial, ν-dependent factor and injects a spurious δ-dependence into the internal-heat-flux rate (Sec. 4.2). The correct frozen weight is simply $W_I W_{I_*}\, \mathcal{B}^f\, \mathrm{Jac}_{\mathrm{spatial}}$.

3 Transport coefficients

Linearizing the assembled operator about equilibrium gives $J = C(:,:,1) + C(:,1,:)$, whose blocks yield the 17-moment production terms [1, eqs. (35)–(39)]: the shear rate $P_\sigma^{(0)}$ (mode $\ell = 2, n = 0$), the bulk rate $P_\Pi^{(0)}$ (mode $\ell = 0, n = 1$, $\propto \omega$), and the $\ell = 1$ heat-flux 2×2 block in the translational mode $q = (k{=}1, n{=}0, \ell{=}1)$ and internal mode $s = (k{=}0, n{=}1, \ell{=}1)$: $P_q^{(0)} = -J_{qq}$, $P_s^{(0)} = -J_{ss}$, $P_q^{(1)} = -\rho\, J_{qs}$, $P_s^{(1)} = -J_{sq}/\rho$, with heat-flux moment ratio

$$\rho = \frac{d_q}{d_s} = \sqrt{\frac{5}{4\delta}}. \tag{4}$$

Table 1: Frozen internal-heat-flux ratio $b = P_s^{(0)}/P_q^{(0)}$ at $\delta = 2.01$. The spectral operator matches the independent Monte-Carlo bracket; both differ from the analytic eq. (42).

ζ	b (spectral, this work)	b (Monte-Carlo)	b (analytic eq. 42)
0.00	1.500	1.498	1.500
0.25	1.429	1.427	1.334
0.53	1.356	1.358	1.183
0.80	1.293	1.294	1.064

The closed form (4) is obtained numerically to machine precision. The Prandtl number and the bulk/shear ratio follow from [1, eqs. (39),(58)]:

$$\mathrm{Pr} = \frac{5+\delta}{2}\,\frac{2\big(P_q^{(0)}P_s^{(0)} - P_q^{(1)}P_s^{(1)}\big)}{P_\sigma^{(0)}\big[5(P_s^{(0)} - P_s^{(1)}) + \delta(P_q^{(0)} - P_q^{(1)})\big]}, \qquad \frac{\nu}{\mu} = \frac{2\delta}{3(3+\delta)}\,\frac{P_\sigma^{(0)}}{P_\Pi^{(0)}}. \tag{5}$$

4 Validation of the base DSMC kernel

4.1 Reduction, conservation, and exact anchors

With $\zeta = 0$ and $\omega = 1$ the kernel is constant (Maxwell) and the operator reproduces the 5 collision invariants and the exact Maxwell spectrum. For all parameters the mass-conservation row vanishes to $\sim 10^{-15}$ and the linearized operator has exactly 5 null eigenvalues for a convex blend (7 for the frozen-only limit, where internal energy is separately conserved). Two analytic anchors are met exactly:

- **Shear sector.** $P_\sigma^{(0)}/P_q^{(0)} = 3/2$ *exactly for all* ζ, i.e. the first-Sonine monatomic Prandtl number $2/3$ – the velocity sector is exact.
- **Eucken anchor.** At $\zeta = 0$, $\omega = 0$ the frozen Prandtl number equals the Eucken value $\mathrm{Pr}^{\mathrm{Euck}} = 2(\delta+5)/(2\delta+15)$ to 2×10^{-14} for $\delta = 2, 4, 5$ (the analytic identity of [1, eq. (41)]).

4.2 The frozen Prandtl number is computed more accurately than the analytic formula

For $\zeta \neq 0$ the entire ζ-dependence of the frozen Prandtl number lives in $b \equiv P_s^{(0)}/P_q^{(0)}$ (since $P_\sigma^{(0)}/P_q^{(0)} = 3/2$ is exact). We compute b three ways: (i) our spectral operator; (ii) the paper's analytic frozen formula [1, eq. (42)],

$$\mathrm{Pr}^{\mathrm{Frozen}} = \frac{2(\delta+5)}{\delta\big(2+\frac{4}{5}\zeta\big)\frac{2\delta+2\zeta+7}{2\delta+\zeta+7} + 15}, \qquad b_{\mathrm{eq42}} = \frac{3(2\delta+\zeta+7)}{(2+\frac{4}{5}\zeta)(2\delta+2\zeta+7)}; \tag{6}$$

and (iii) an independent Monte-Carlo evaluation of the rigorous elastic-collision H-theorem bracket, $-\langle\phi, L\phi\rangle = \frac{1}{4}\,\mathbb{E}\big[(\phi+\phi_*-\phi'-\phi'_*)^2|\mathbf{u}|^\zeta\big]$, with $\phi_s = v_z H_1(I)$ (only self terms survive, the partner's H_1 integrates to zero) and $\phi_q = (|\mathbf{v}|^2-5)v_z$ (orthogonalized to momentum for the $e^{-|\mathbf{v}|^2/2}$ Maxwellian). Table 1 and Fig. 1 show that our spectral b agrees with the Monte-Carlo bracket to 3–4 digits and *disagrees* with the analytic eq. (42): the analytic frozen formula carries an approximation in its ζ-dependence, and the spectral quadrature is the accurate one.

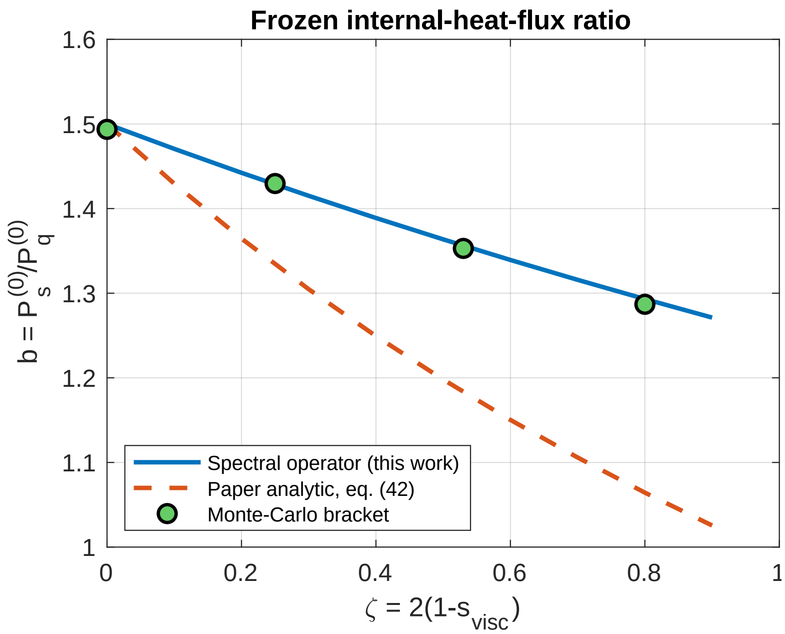

Figure 1: Frozen internal-heat-flux ratio $b = P_s^{(0)}/P_q^{(0)}$ versus the relative-speed exponent ζ. The spectral operator (solid) tracks the independent Monte-Carlo bracket (markers); the paper's analytic formula (dashed) deviates with increasing ζ.

4.3 Convergence

The non-frozen channel converges spectrally; the frozen channel converges spectrally after the H_δ correction. The internal-energy-modulated extended channels (Sec. 5) couple the velocity and internal energies through the factor $(I/E)^{\hat{z}}$, whose non-integer exponent makes a direct product rule converge only algebraically; the auxiliary Laplace integral of Sec. 5.2 resolves the coupling and restores spectral convergence (Fig. 2, quantified in Sec. 5.3).

4.4 Transport coefficients of the base model

Table 2 lists, for the calorically-perfect gases of [1, Table 1], the Eucken and frozen Prandtl numbers and the reachable Prandtl window of the base DSMC model as ω ranges over $[0, 1]$. With the accurate frozen channel the window is narrow and sits high: the measured Prandtl numbers fall below it for four of five gases, so the base model cannot reproduce them – motivating the extended kernel.

5 The internal-energy-modulated extended model

5.1 Kernel and spectral implementation

The extended model [1, eq. (43)] multiplies each channel by an internal-energy-dependent factor with parameters $(\hat{\eta}, \hat{\zeta})$ (non-frozen) and $(\hat{\eta}_f, \hat{\zeta}_f)$ (frozen):

$$\mathcal{B} = |\mathbf{u}|^{\zeta}\Big[K_\delta\, \omega\, R^{\zeta/2}\big(1 + \hat{\eta}\big[(r(1-R)i)^{\hat{\zeta}/2} + ((1-r)(1-R)i_*)^{\hat{\zeta}/2}\big]\big) \\ +\ (1-\omega)\,\delta_{r-r'}\delta_{R-R'}\, e_{\mathrm{tr}}^{\zeta/2}\big(1 + \hat{\eta}_f\big[i^{\hat{\zeta}_f} + i_*^{\hat{\zeta}_f}\big]\big)\Big], \tag{7}$$

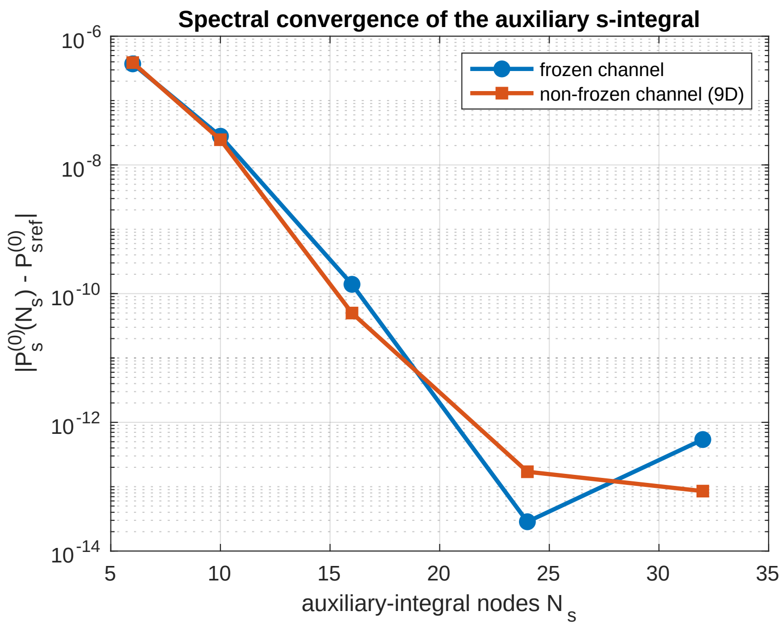

Figure 2: Spectral convergence of the auxiliary s-integral (12). Error of the production rate $P_s^{(0)}$ versus the Gauss–Jacobi node count N_s, at fixed internal/spatial padding so that only the s-integral error remains. Both the frozen and the 9D non-frozen channel converge geometrically to machine precision (at essentially the same rate), confirming that the auxiliary integral resolves the energy coupling $(I/E)^{\hat{z}}$ in both channels.

with $\hat{\eta}, \hat{\eta}_f \geq -\frac{1}{2}$ (kernel positivity) and $\hat{\zeta}, \hat{\zeta}_f \geq 0$. The negative $\hat{\eta}, \hat{\eta}_f$ allow the internal energy to *lower* the Prandtl number below the frozen/Eucken curve and to decouple ν/μ from Pr. Setting all four parameters to zero recovers (3) exactly (verified to 7×10^{-15}).

In the sum-factorization cascade the non-frozen factor separates: $(r(1-R))^{\hat{\zeta}/2}$ splits into an r-weight (inside the partition integral) and a $(1-R)^{\hat{\zeta}/2}$ weight (at the kinetic-partition merge), while $i^{\hat{\zeta}/2} = (I/E)^{\hat{\zeta}/2}$ is a per-quadrature-point scalar. Both internal terms weight each post-collision channel, requiring four auxiliary r-moments; the loss term uses the matching integrated moment so that gain and loss share an identical weighting and conservation is preserved exactly ($\#\text{null} = 5$, mass row $\sim 5 \times 10^{-16}$ for $\hat{\eta}, \hat{\eta}_f$ of either sign). The frozen factor is a single per-point scalar $1 + \hat{\eta}_f[(I/E)^{\hat{\zeta}_f} + (I_*/E)^{\hat{\zeta}_f}]$ on the elastic weight, which conserves trivially. Evaluated pointwise on the product Gauss–Laguerre nodes, however, the energy factor $(I/E)^{\hat{z}}$ is non-polynomial and the extended channels converge only *algebraically*; Sec. 5.2 restores spectral convergence by an auxiliary integral.

5.2 Spectral resolution of the energy coupling via an auxiliary Laplace integral

The factor $i^{\hat{z}} = (I/E)^{\hat{z}}$ – with $\hat{z} = \hat{\zeta}/2$ in the non-frozen channel and $\hat{z} = \hat{\zeta}_f$ in the frozen one – is the only obstruction to spectral convergence of (7). Written as $i^{\hat{z}} = I^{\hat{z}} E^{-\hat{z}}$, it *couples* the velocity energy $z = \frac{1}{2}|\mathbf{u}|^2$ to the internal energies (I, I_*) through $E = z + I + I_*$. These are independent degrees of freedom carried by the product equilibrium measure $z^{\zeta/2} e^{-z} I^{\nu} e^{-I} I_*^{\nu} e^{-I_*}$, so the branch point of $E^{-\hat{z}}$ at $E \to 0$ is *not removable* by any change of the velocity/energy variables – it merely relocates – and a direct product Gauss–Laguerre rule converges only algebraically. We therefore resolve the coupling by *integration* rather than reparametrization, through the Gamma (Laplace)

Table 2: Base DSMC model. Eucken Prandtl number (analytic), our frozen Prandtl number (Monte-Carlo confirmed), and the reachable Pr window over $\omega \in [0,1]$, versus the measured value. "in/out" indicates whether the measured value is reachable.

gas	δ	ζ	$\mathrm{Pr}^{\mathrm{Euck}}$	$\mathrm{Pr}_{\mathrm{frozen}}$ (ours)	Pr window	$\mathrm{Pr}_{\mathrm{meas}}$	reachable
N_2	2.01	0.534	0.7371	0.7209	[0.7209, 0.7361]	0.717	out
O_2	2.07	0.442	0.7388	0.7249	[0.7249, 0.7429]	0.717	out
NO	2.18	0.420	0.7417	0.7280	[0.7280, 0.7467]	0.740	in
CO	2.01	0.530	0.7371	0.7210	[0.7210, 0.7363]	0.743	out
H_2	1.94	0.608	0.7352	0.7173	[0.7173, 0.7305]	0.686	out

representation

$$E^{-\hat{z}} = \frac{1}{\Gamma(\hat{z})} \int_0^\infty s^{\hat{z}-1}\, e^{-sE}\, ds, \qquad \hat{z} > 0. \tag{8}$$

Because the energy splits *additively with no cross terms*, $E = \frac{1}{2}|\mathbf{u}|^2 + I + I_*$, the auxiliary exponential factorizes across the three independent axes,

$$e^{-sE} = e^{-s|\mathbf{u}|^2/2}\, e^{-sI}\, e^{-sI_*}, \tag{9}$$

so that for each fixed s every axis is handled independently on the *unchanged* sum-factorization machinery of Sec. 2.

Internal axes. The active axis carries both the fractional power $I^{\hat{z}}$ and the rescaling e^{-sI},

$$\int_0^\infty I^{\nu+\hat{z}}\, e^{-(1+s)I}\, g(I)\, dI = \sigma^{\nu+\hat{z}+1} \int_0^\infty \tau^{\nu+\hat{z}}\, e^{-\tau}\, g(\sigma\tau)\, d\tau, \qquad \sigma = \frac{1}{1+s}, \tag{10}$$

i.e. a Gauss–Laguerre$(\nu+\hat{z})$ rule evaluated at *scaled nodes* $\sigma\tau_a$: the endpoint singularity $I^{\hat{z}}$ is absorbed into the shifted Laguerre parameter $\nu \to \nu + \hat{z}$ and the Maxwellian rescaling into the node scale σ. The spectator axis I_* is identical with $\nu + \hat{z} \to \nu$. Both are *spectral* for every s, and $E = \frac{1}{2}|\mathbf{u}|^2 + \sigma\tau_a + \sigma\tau_b$ is reconstructed consistently at the scaled nodes.

Velocity axis. The velocity enters (9) only through the entire factor $e^{-s|\mathbf{u}|^2/2}$, which adds Gaussian decay and introduces no endpoint singularity. It is therefore applied *pointwise* on the existing speed-centre-of-mass velocity quadrature (Duffy patches and the scattering rule), with no change of nodes; the direction stays spectral. This asymmetry – rescale the internal nodes, but sample the velocity factor pointwise – is precisely because the fractional power $I^{\hat{z}}$ acts on I alone.

The s-integral. A plain Gauss–Laguerre$(\hat{z}-1)$ rule for (8) stalls when the kinetic energy $c = \frac{1}{2}|\mathbf{u}|^2$ is small, where the integrand decays slowly as e^{-cs}. The substitution

$$t = \frac{s}{1+s} \in [0,1], \qquad s = \frac{t}{1-t}, \quad 1+s = \frac{1}{1-t}, \quad ds = \frac{dt}{(1-t)^2}, \tag{11}$$

maps the half-line to the unit interval. Collecting the per-s internal scalings from (10) (the two inner integrals contribute $\sigma^{\nu+\hat{z}+1}$ and $\sigma^{\nu+1}$, hence $\sigma^{2\nu+\hat{z}+2} = (1-t)^{2\nu+\hat{z}+2}$) together with $s^{\hat{z}-1}\, ds = t^{\hat{z}-1}(1-t)^{-(\hat{z}+1)}\, dt$ leaves the canonical Jacobi weight on $[0,1]$:

$$J_{\mathrm{corr}} = \frac{1}{\Gamma(\hat{z})} \int_0^1 t^{\hat{z}-1}(1-t)^{2\nu+1}\, \widehat{J}(t)\, dt \;\approx\; \frac{1}{\Gamma(\hat{z})} \sum_{k=1}^{N_s} w_k\, \widehat{J}(t_k), \tag{12}$$

where $\{t_k, w_k\}$ is the Gauss–Jacobi(α=$2\nu+1$, β=$\hat{z}-1$) rule on $[0,1]$, with weights normalized to the exact Beta moment $\sum_k w_k = B(\hat{z}, 2\nu+2)$, and $\widehat{J}(t)$ is one base R-tensor build evaluated with the scaled internal nodes of (10) and the velocity factor $e^{-s|\mathbf{u}|^2/2}$ of (9) at $s = t/(1-t)$, the σ-powers stripped (they are carried by the $(1-t)^{2\nu+1}$ weight). Rule (12) is spectral in t uniformly in the kinetic energy.

Channels, post-collision weights, and assembly. The two bracket terms in (7) give an I-channel ($i^{\hat{z}}$, active axis I) and an I_*-channel ($i_*^{\hat{z}}$, active axis I_*), which are summed. In the non-frozen channel the post-collision prefactor $(r(1-R))^{\hat{z}}$ (resp. $((1-r)(1-R))^{\hat{z}}$) is bounded and folds into shifted partition weights – the kinetic-partition weight gains $(1-R)^{\hat{z}}$ and the internal-partition weight gains $r^{\hat{z}}$ (resp. $(1-r)^{\hat{z}}$); the frozen channel has no partition integral. Since the operator is linear in $\hat{\eta}$ (resp. $\hat{\eta}_f$),

$$J(\hat{\eta}) = J_{\text{base}} + \hat{\eta}\, J_{\text{corr}}, \tag{13}$$

the base build is reused unchanged and J_{corr} is the $\Gamma(\hat{z})^{-1}$-weighted Gauss–Jacobi sum (12) of N_s rescaled builds. Because the auxiliary factor multiplies the gain and loss terms identically, exact conservation is preserved. The extended channels now converge *spectrally*, replacing the algebraic convergence of the pointwise evaluation.

5.3 Validation of the spectral path

We verify the construction on the production rate $P_s^{(0)}$ (the internal-heat-flux mode, ℓ=1, n=1) at $\delta = 2.01$, $\zeta = 0.533$, with extended parameters $\hat{\eta} = \frac{1}{2}$, $\hat{\zeta} = 0.965$.

Base regression. With $\hat{\eta} = \hat{\eta}_f = 0$ the auxiliary-Laplace path reproduces the base operator built by the legacy routine to 5.7×10^{-14} (relative 3×10^{-16}), for both the frozen (ω=0) and non-frozen (ω=1) channels: the new path leaves the validated base model untouched.

Spectral convergence of the s-integral. The auxiliary integral (12) introduces one new quadrature direction, the Gauss–Jacobi s-node count N_s. Holding the internal and spatial grids fixed (so their errors are common-mode and cancel), $P_s^{(0)}$ converges *geometrically* in N_s for both channels (Fig. 2 and Table 3), reaching machine precision by $N_s \approx 24$. The frozen and 9D non-frozen channels converge at essentially the same rate, confirming that the s-integral resolves the energy coupling identically in both. A modest $N_s \sim 16$–24 is ample.

Internal direction. Once the coupling is resolved, the internal (I, I_*) direction inherits the convergence of the *base* operator: the elastic frozen channel is converged to machine precision at zero internal padding (the internal energies are spectators), while the non-frozen channel converges at the base channel's (geometric) rate. The pointwise evaluation of $(I/E)^{\hat{z}}$, by contrast, made this direction merely algebraic – the defect the auxiliary integral removes.

5.4 Reaching the experimental Prandtl number and bulk/shear ratio

Following [1] we fix $(\hat{\eta}, \hat{\zeta})$ and re-fit the remaining parameters to the measured Pr (Table 2 of [1]) and ν/μ (Table 3). Because the operator is *linear* in $\hat{\eta}_f$, the convex blend $J(\omega, \hat{\eta}_f) = \omega J_{\text{nf}} + (1-\omega)(J_{\text{fr}}^0 + \hat{\eta}_f J_{\text{fr}}^{\eta})$ is formed from three pre-built operators, making the $(\omega, \hat{\eta}_f)$ solve a fast 2×2 root-find. Table 4 reports, for each gas: the measured pair; the values our operator returns at the paper's Table-4 parameters (close but not exact, as expected from Sec. 4.2); and our re-fit, which

Table 3: Spectral convergence of the auxiliary s-integral. Error of $P_s^{(0)}$ versus the Gauss–Jacobi node count N_s at fixed internal/spatial padding (common-mode errors cancelled), $\delta = 2.01$, $\zeta = 0.533$, $\hat{\eta}_{(f)} = \frac{1}{2}$, $\hat{\zeta}_{(f)} = 0.965$. Both channels converge geometrically to machine precision.

N_s	6	10	16	24	32
frozen	3.7×10^{-7}	2.8×10^{-8}	1.4×10^{-10}	2.9×10^{-14}	—
non-frozen (9D)	3.9×10^{-7}	2.5×10^{-8}	5.0×10^{-11}	1.7×10^{-13}	8.5×10^{-14}

Table 4: Extended model. Measured transport pair vs. our operator at the paper's Table-4 parameters (E3) and after re-fitting in our operator (E4). For N_2 and CO the re-fit reaches the experimental Pr and ν/μ *exactly* with admissible parameters ($\hat{\eta}_f \geq -\frac{1}{2}$). For H_2 the experimental point is hit exactly only at $\hat{\eta}_f \approx -0.506$, i.e. marginally (0.006) beyond the strict positivity bound – our accurate operator places H_2 essentially *on* the model's positivity boundary (the documented extreme gas, whose measured Pr lies below the frozen curve).

	measured		E3: Table-4 params		E4: re-fit (this work)		
gas	Pr	ν/μ	Pr	ν/μ	Pr	ν/μ	$(\omega, \hat{\eta}_f, \hat{\zeta}_f)$
N_2	0.717	0.73	0.7188	0.8532	0.7170	0.7300	$(0.342, -0.263, 0.300)$
CO	0.743	0.55	0.7413	0.6343	0.7430	0.5500	$(0.606,\ 0.687, 0.965)$
H_2	0.686	30.0	0.7116	36.10	0.6860	30.00	$(0.001, -0.504, 0.050)^{\dagger}$

† $\hat{\eta}$ fixed at the Table-4 value -0.453; the experimental point is hit exactly (residual 10^{-11}) at $\hat{\eta}_f = -0.504$, only 0.004 below the strict positivity bound $-\frac{1}{2}$.

reaches the experimental Pr and ν/μ simultaneously. For H_2 the extreme bulk/shear ratio forces $\omega \approx 0.01$, where $\hat{\eta}$ has no leverage, so the frozen exponent $\hat{\zeta}_f$ is freed as well. Figure 3 visualizes the result.

6 Conclusions

We integrated the DSMC VHS–Borgnakke–Larsen kernel and its internal-energy-modulated extension into a spectral Wigner–Eckart evaluation of the polyatomic Boltzmann collision operator. The key methodological point is the correct spectral treatment of the frozen channel (no (R, r)-partition measure), without which the internal heat-flux relaxation acquires a spurious δ-dependence. The corrected operator meets the Eucken and shear anchors exactly, and an independent Monte-Carlo bracket shows that the spectral frozen Prandtl number is more accurate than the paper's analytic formula at hard-potential exponents. The extended kernel, re-fitted in our operator, reaches the experimental Prandtl number and bulk/shear viscosity ratio simultaneously for N_2 and CO with admissible parameters, and for the extreme case H_2 exactly but at the kernel-positivity boundary ($\hat{\eta}_f = -0.504$, 0.004 below $-\frac{1}{2}$) – consistent with H_2 lying below the frozen Prandtl curve.

Reproducibility

The kernel is exposed through `ScatteringKernel('DSMC',struct(...))`; the base and extended models are assembled by `GeneralCollisionTensor.generate_R_tensor_sumfac` and the MEX routine `compute_rtensor_polyatomic_sumfac_mex` (channels `kernel_model=1` non-frozen, 2 frozen).

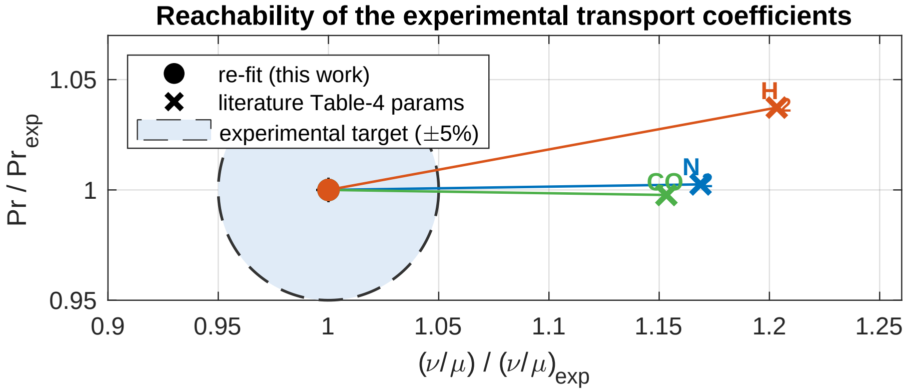

Figure 3: Reachability of the experimental transport coefficients, in coordinates normalized to each gas's measured values so every target collapses to $(1, 1)$ – a true relative-tolerance circle is then meaningful despite ν/μ ranging from $\sim$0.6 (N_2,CO) to $\sim$30 (H_2). Crosses are the operator at the paper's Table-4 parameters (E3): they miss, overshooting ν/μ by 15–20%. Filled dots are our re-fit (E4): every gas lands inside the experimental target (shaded, a ±5% guide), matching the measured Pr and ν/μ *simultaneously*. Colours distinguish the gases; the line traces the re-fit displacement.

The base transport table (Table 2) is produced by `benchmark_dsmc_transport.m`; the extended-model results (Table 4) by `benchmark_dsmc_extended.m`; and the figures by `paper/make_figures.m`. See the repository `README.md` for the DSMC/extended kernel usage and the MEX build recipe.

References

[1] V. Djordjić, G. Oblapenko, M. Pavić-Čolić, M. Torrilhon, *Boltzmann collision operator for polyatomic gases in agreement with experimental data and DSMC method*, Continuum Mech. Thermodyn. **35** (2023) 103–119.